

CX4240 Spring 2026

Unsupervised Learning: Gaussian Mixture Models

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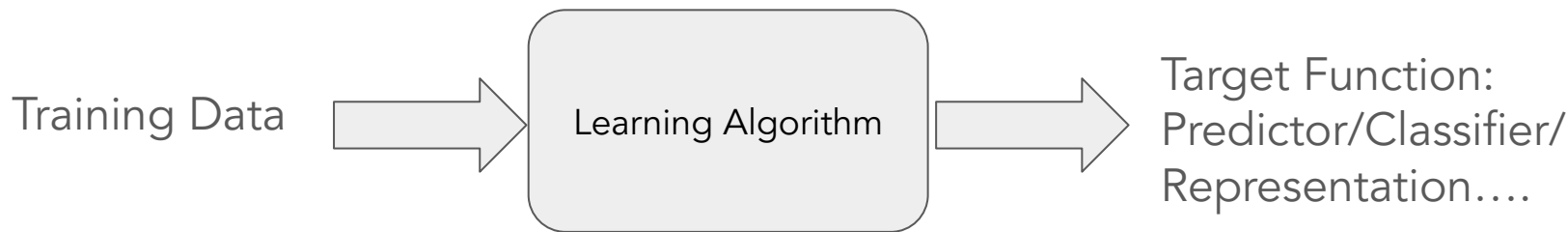
Grading Policy Update

- Based on the poll on Ed, we will have an additional proposal with 10% bonus credit.
- The deadline will be on April 13th.
- The practice exam will be released on Mar 6th.
- We will have a makeup exam on March 16th or 17th (TBD)
 - You need an approval of absence for this makeup exam

Proposal Requirement

- All write-ups should use the [NeurIPS style](#).
- Your final report is expected to be up to **3 pages** excluding references. It should have roughly the following format:
 - Introduction: Problem definition and motivation
 - Background & Related Work: Background info and literature survey

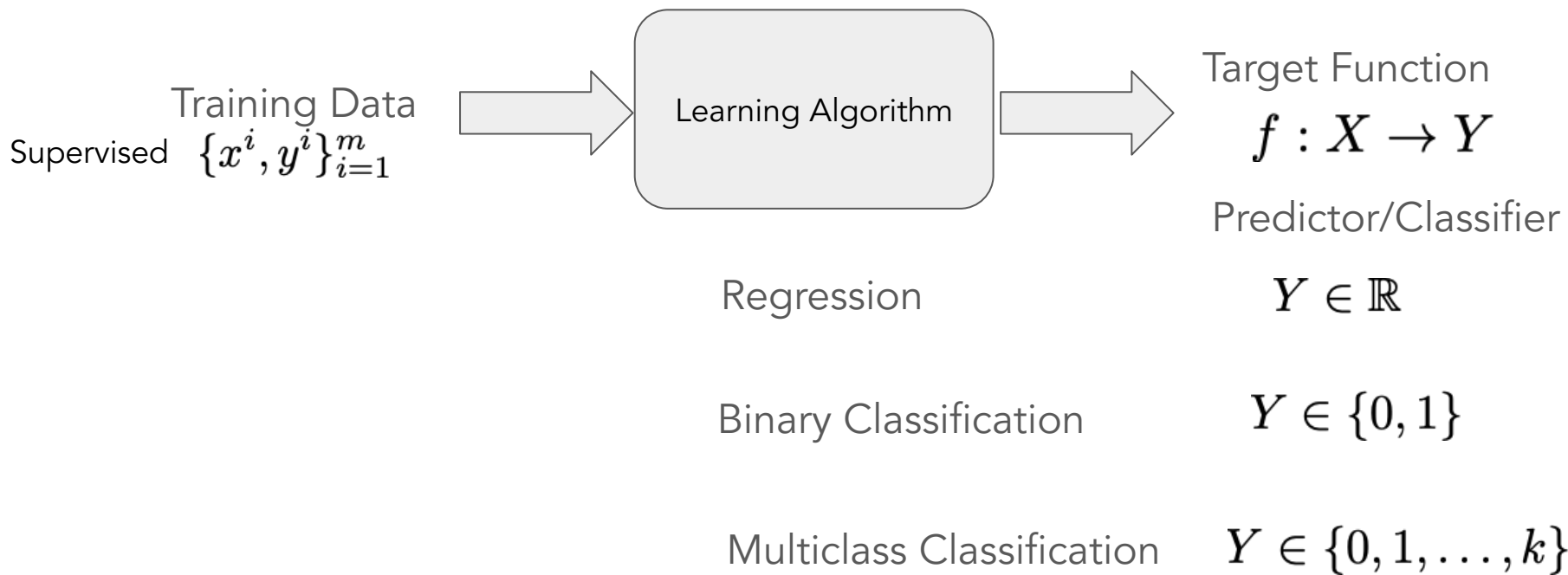
ML Algorithm Pipeline



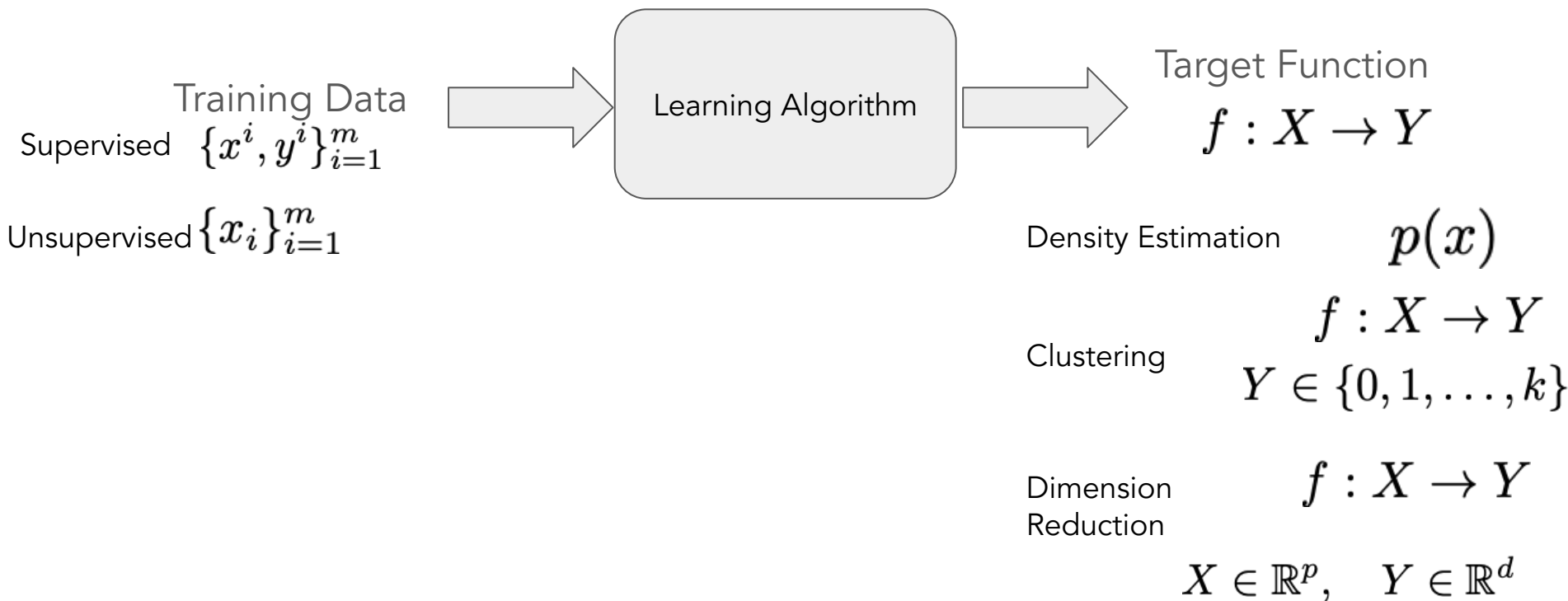
General ML Algorithm Pipeline

1. Build probabilistic models
2. Derive loss function (by MLE or MAP....)
3. Select optimizer

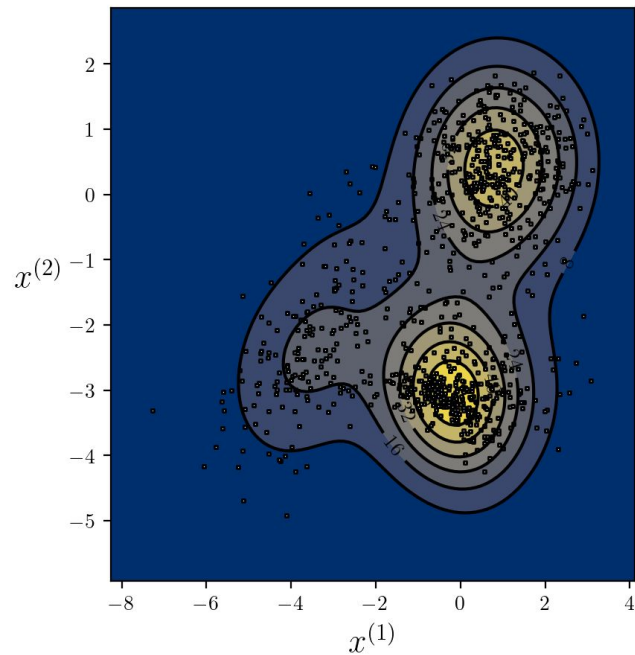
Supervised Learning vs. Unsupervised Learning



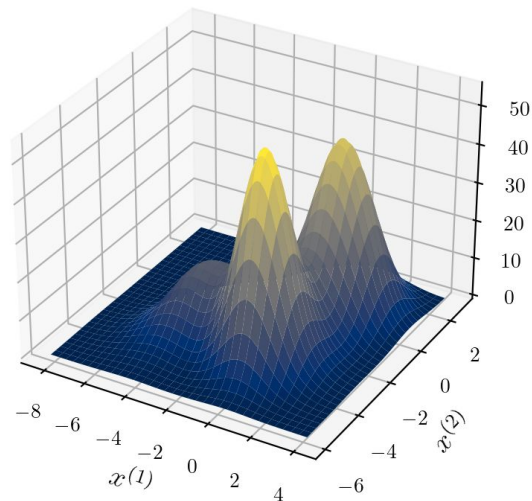
Supervised Learning vs. Unsupervised Learning



Density Estimation

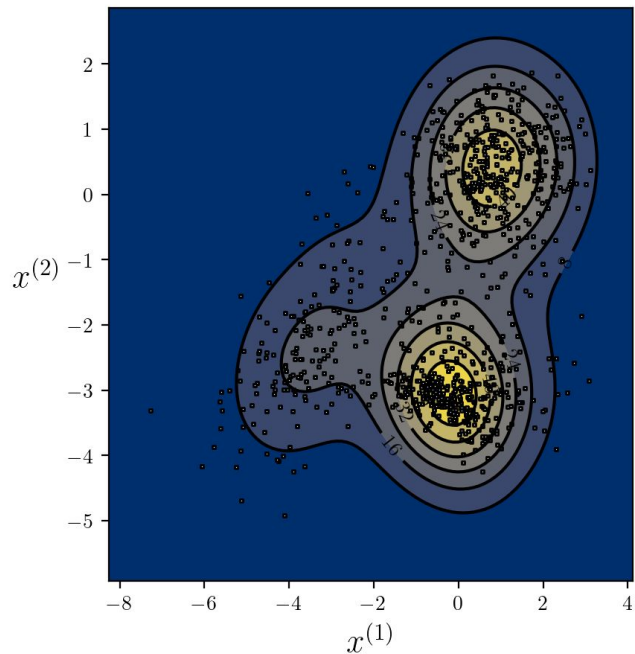


$$\{x_i\}_{i=1}^m$$

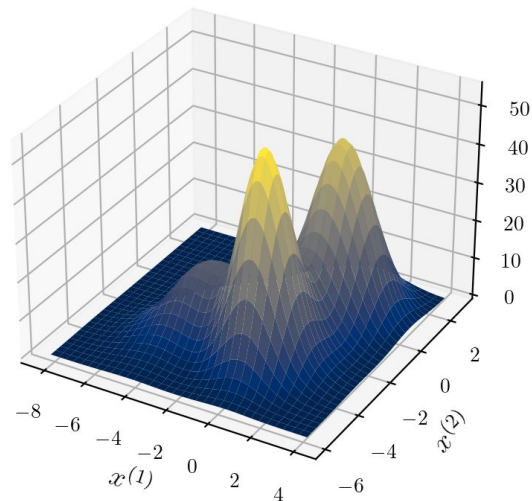


$$p(x)$$

Density Estimation



$$\{x_i\}_{i=1}^m$$



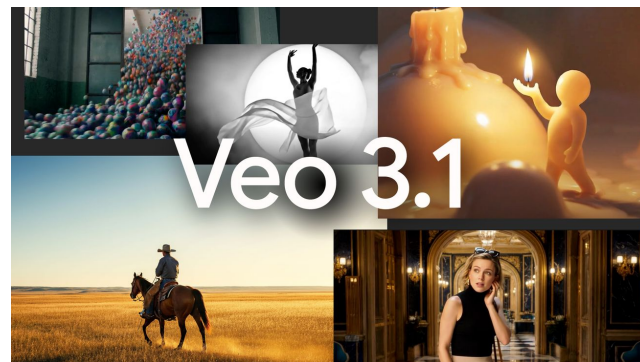
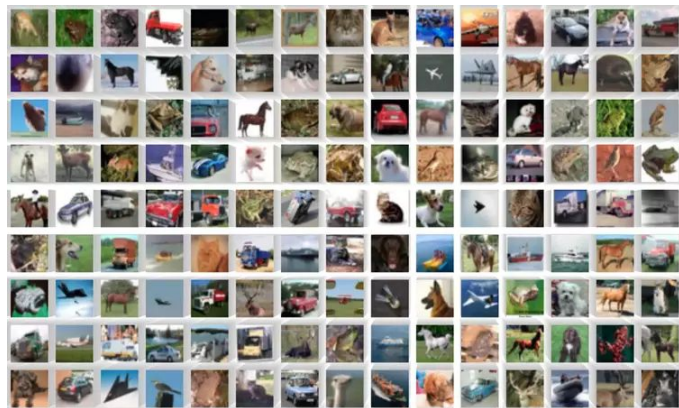
$$p(x)$$

Generative Models

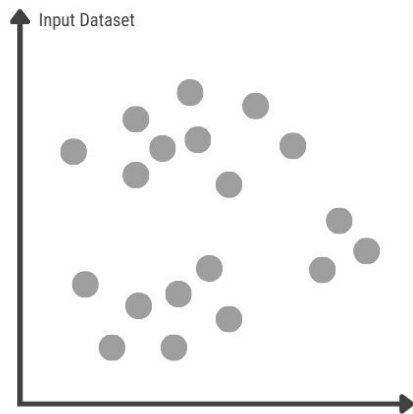
$$x \sim p(x)$$

Density Estimation: Generative Models

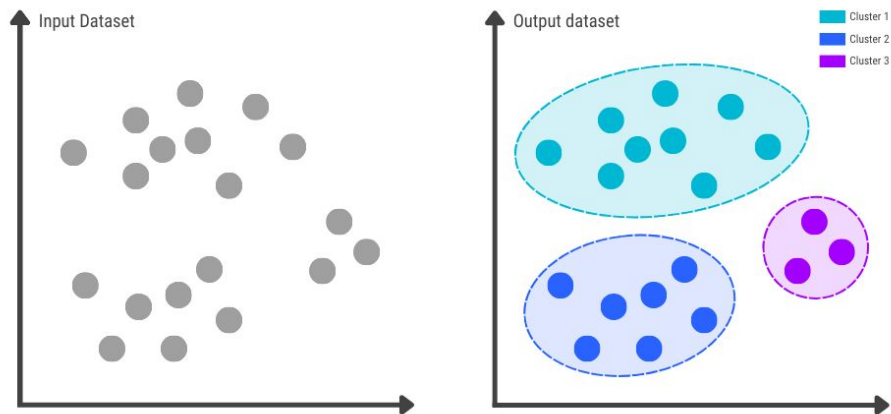
$$x \sim p(x)$$



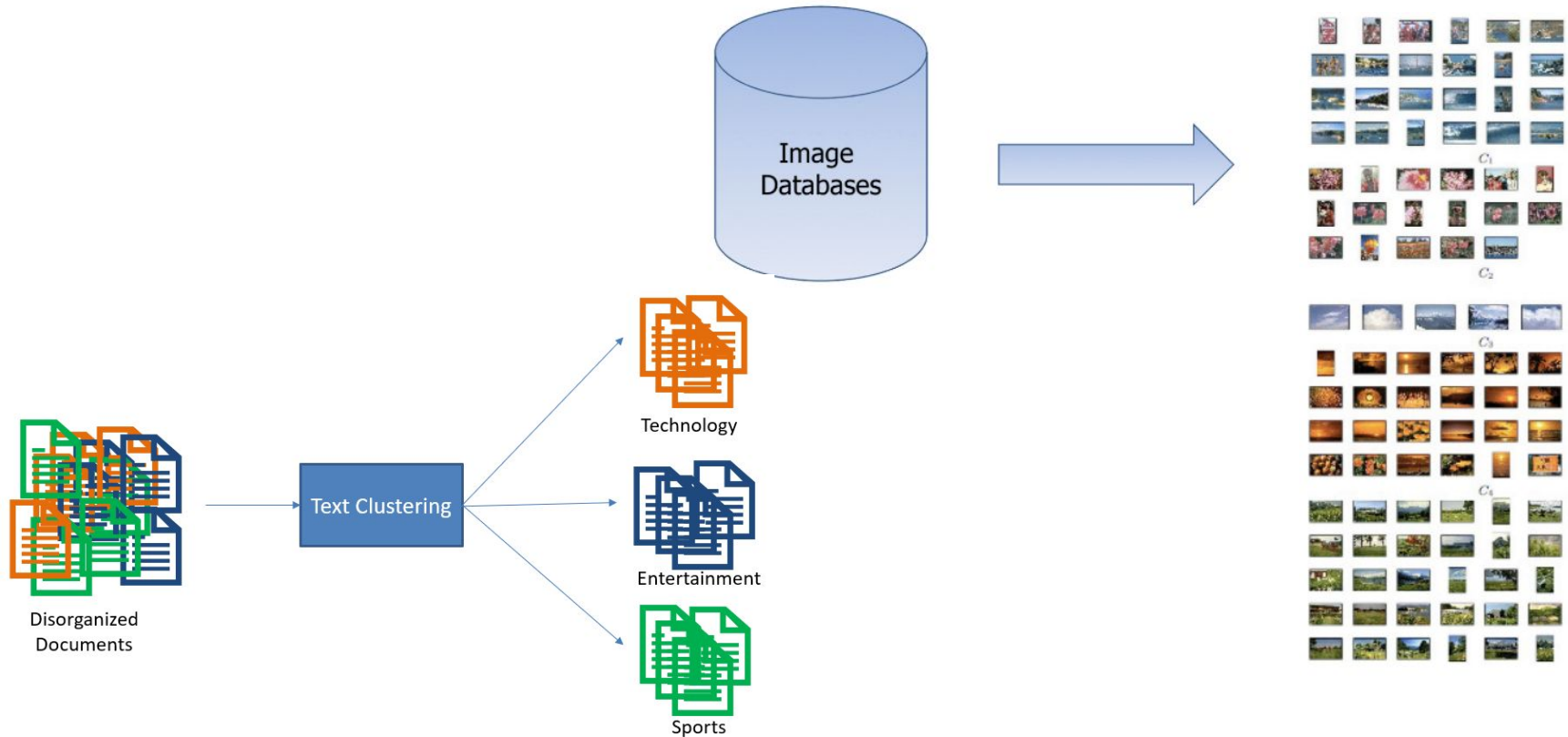
Clustering



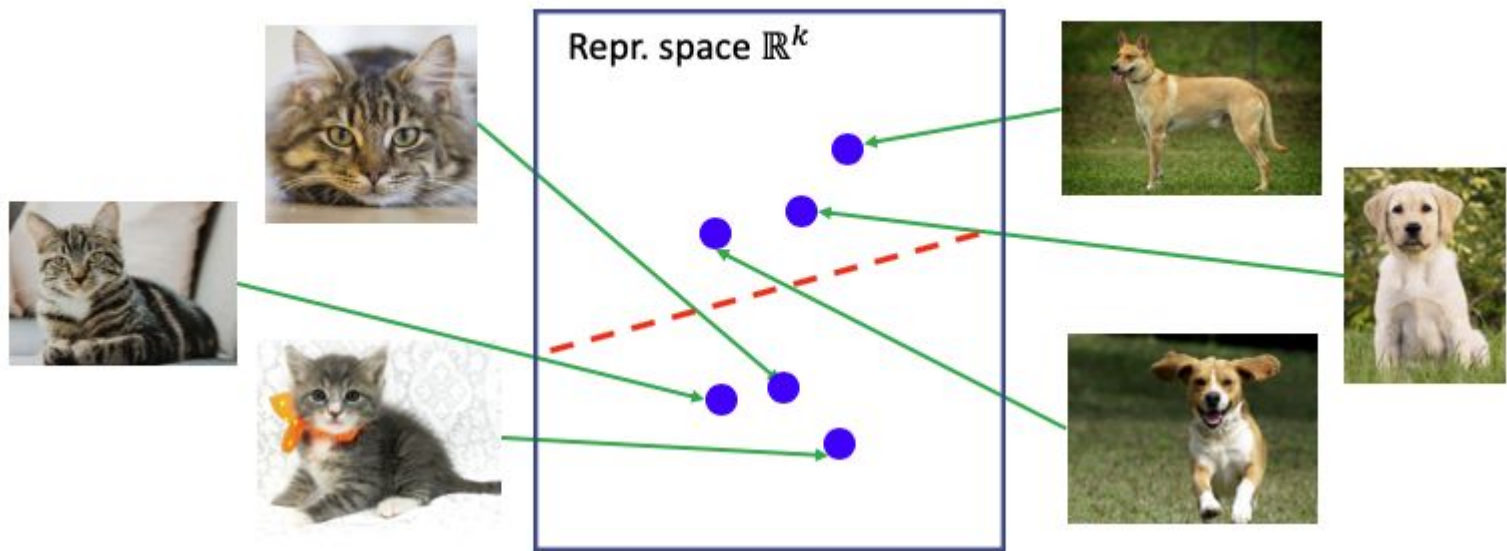
Clustering



Clustering: Data Organization



Dimension Reduction/Representation Learning



Density Estimation



Density Estimation

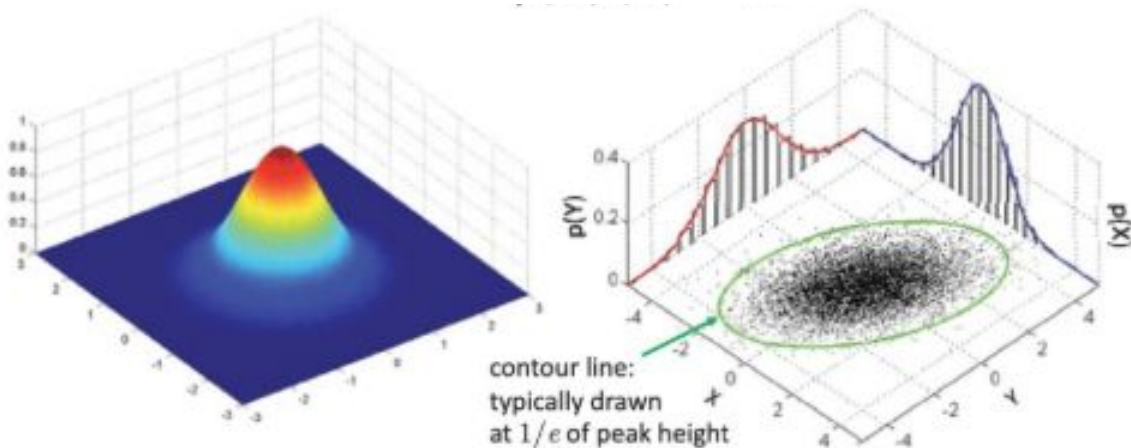


Density Estimation Pipeline

1. Build probabilistic models
2. Derive loss function (by MLE or MAP....)
3. Select optimizer

Gaussian Distribution

$$X_i \sim \mathcal{N}(\mu, \Sigma) \quad p(x_i; \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp \left(-\frac{1}{2} (x_i - \mu)^T \Sigma^{-1} (x_i - \mu) \right)$$



Density Estimation



Density Estimation Pipeline

1. Build probabilistic models
2. Derive loss function (by MLE or MAP...)
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MLE of Gaussian Model

The log-likelihood is given by

$$\log L(\mu, \Sigma) = \sum_{i=1}^m \log p(x_i | \mu, \Sigma) = -\frac{mn}{2} \log(2\pi) - \frac{m}{2} \log |\Sigma| - \frac{1}{2} \sum_{i=1}^m (x_i - \mu)^T \Sigma^{-1} (x_i - \mu)$$

The maximum likelihood estimator is

$$(\hat{\mu}, \hat{\Sigma}) = \arg \max_{\mu, \Sigma} \log p(\mathbf{X} | \mu, \Sigma)$$

Density Estimation



Density Estimation Pipeline

1. Build probabilistic models
2. Derive loss function (by MLE or MAP....)
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Gradient Calculation of MLE

$$\nabla_{\mu} \log p(\mathbf{X} \mid \mu, \Sigma) = \Sigma^{-1} \sum_{i=1}^m (\mathbf{x}_i - \mu)$$

$$\nabla_{\Sigma} \log p(\mathbf{X} \mid \mu, \Sigma) = -\frac{m}{2} \Sigma^{-1} + \frac{1}{2} \sum_{i=1}^m \Sigma^{-1} (\mathbf{x}_i - \mu) (\mathbf{x}_i - \mu)^T \Sigma^{-1}$$

The closed-form solution is

$$\hat{\mu}_{\text{MLE}} = \frac{1}{m} \sum_{i=1}^m \mathbf{x}_i \quad \hat{\Sigma}_{\text{MLE}} = \frac{1}{m} \sum_{i=1}^m \left(\mathbf{x}_i - \hat{\mu}_{\text{MLE}} \right) \left(\mathbf{x}_i - \hat{\mu}_{\text{MLE}} \right)^T$$

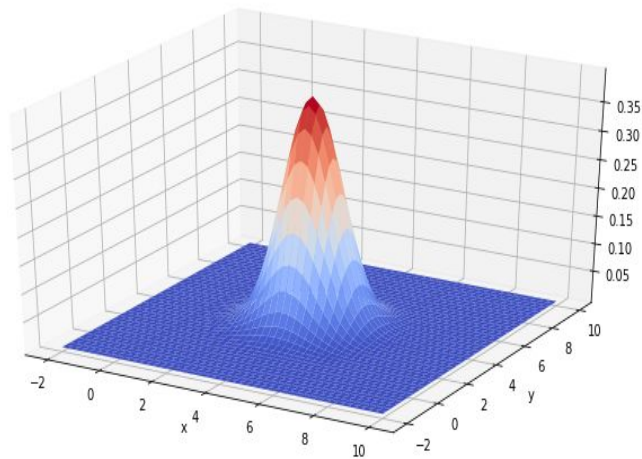
Density Estimation: Gaussian Model



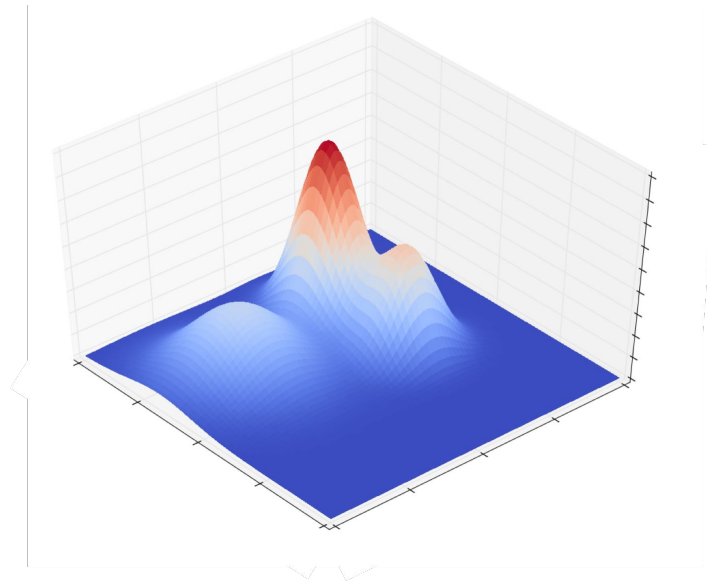
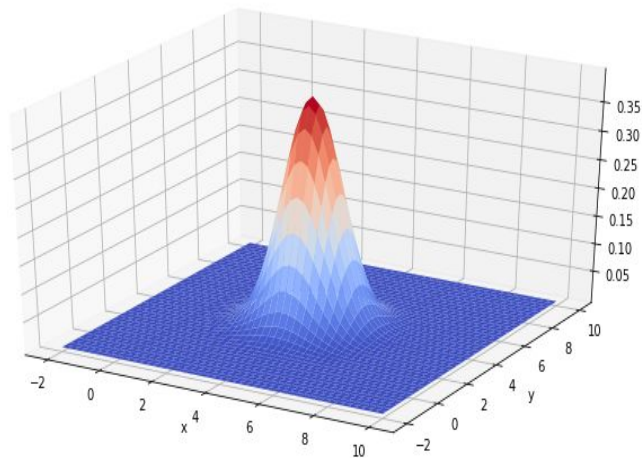
Density Estimation Pipeline

1. Build probabilistic models
Gaussian Distribution
2. Derive loss function (by MLE or MAP....)
MLE
3. Select optimizer
Necessary Condition

Mixture of Gaussians



Mixture of Gaussians



Density Estimation



Density Estimation Pipeline

1. Build probabilistic models
Gaussian Mixture Models
2. Derive loss function (by MLE or MAP....)
3. Select optimizer

Gaussian Mixture Model

Class mixture prior: $P(y)$ $\pi = (\pi_1, \pi_2, \dots, \pi_k), \sum_{i=1}^k \pi_i = 1, \pi_i \geq 0$

Class conditional distribution: $p(x|y) = \mathcal{N}(x|\mu_y, \Sigma_y)$

Marginal distribution: $P(x) = \sum_y P(x|y)P(y) = \sum_{i=1}^k \pi_i \mathcal{N}(x|\mu_i, \Sigma_i)$

Connection to Gaussian Naive Bayes

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)} = \frac{P(x, y)}{\sum_z P(x, y)}$$

Prior: $P(y)$ $\pi = (\pi_1, \pi_2, \dots, \pi_k)$, $\sum_{i=1}^k \pi_i = 1, \pi_i \geq 0$

Likelihood (class conditional distribution): $p(x|y) = \mathcal{N}(x|\mu_y, \Sigma_y)$

Posterior: $P(y|x) = \frac{P(y)\mathcal{N}(x|\mu_y, \Sigma_y)}{\sum_y P(y)\mathcal{N}(x|\mu_y, \Sigma_y)}$

Connection to Gaussian Naive Bayes

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)} = \frac{P(x, y)}{\sum_z P(x, y)}$$

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What is the difference?

Density Estimation



Density Estimation Pipeline

1. Build probabilistic models
Gaussian Mixture Models
2. Derive loss function (by MLE or MAP....)
3. Select optimizer

MLE of GMM

$$\max_{\{\pi_i, \mu_i, \Sigma_i\}_{i=1}^k} \sum_{j=1}^m \log p(x_j) = \sum_{j=1}^m \log \left(\sum_{i=1}^k \pi_i \mathcal{N}(x_j | \mu_i, \Sigma_i) \right)$$

Want $\arg \max_{\theta} \log L(\theta)$ subject to $\sum_{j=1}^k \pi_j = 1$

MLE of GMM vs. MLE of Gaussian Naive Bayes

$$\max_{\{\pi_i, \mu_i, \Sigma_i\}_{i=1}^k} \sum_{j=1}^m \log p(x_j) = \sum_{j=1}^m \log \left(\sum_{i=1}^k \pi_i \mathcal{N}(x_j | \mu_i, \Sigma_i) \right)$$

$$\max_{\{\pi_i, \mu_i, \Sigma_i\}_{i=1}^k} \sum_{j=1}^m \log p(x_j, y_j) = \sum_{j=1}^m \log \pi_{y_j} + \log \mathcal{N}(x_j | \mu_{y_j}, \Sigma_{y_j})$$

Want $\arg \max_{\theta} \log L(\theta)$ subject to $\sum_{j=1}^k \pi_j = 1$

Density Estimation



Density Estimation Pipeline

1. Build probabilistic models
Gaussian Mixture Models
2. Derive loss function (by MLE or MAP....)
3. Select optimizer

Select Optimizer

Stochastic Gradient?

$$\max_{\{\pi_i, \mu_i, \Sigma_i\}_{i=1}^k} \sum_{j=1}^m \log p(x_j) = \sum_{j=1}^m \log \left(\sum_{i=1}^k \pi_i \mathcal{N}(x_j | \mu_i, \Sigma_i) \right)$$

Plausible but tedious

If label is given....Gaussian Naive Bayes!

$$\max_{\{\pi_i, \mu_i, \Sigma_i\}_{i=1}^k} \sum_{j=1}^m \log p(x_j) = \sum_{j=1}^m \log \left(\sum_{i=1}^k \pi_i \mathcal{N}(x_j | \mu_i, \Sigma_i) \right)$$

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Easy to solve

Want $\arg \max_{\theta} \log L(\theta)$ subject to $\sum_{j=1}^k \pi_j = 1$

If label is given....Gaussian Naive Bayes!

$$\log L(\theta) = \sum_{i=1}^m \sum_{j=1}^k y_j^i \log \pi_j - \sum_{i=1}^m \log Z - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^k y_j^i (x^i - \mu_j)^\top \Sigma_j^{-1} (x^i - \mu_j)$$

$$\max_{\{\pi_i, \mu_i, \Sigma_i\}_{i=1}^k} \sum_{j=1}^m \log p(x_j, y_j) = \sum_{j=1}^m \log \pi_{y_j} + \log \mathcal{N}(x_j | \mu_{y_j}, \Sigma_{y_j})$$

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$$\frac{\partial \log L}{\partial \mu_k} = - \sum_{i=1}^m y_k^i \Sigma_k^{-1} (x^i - \mu_k) = 0$$

$$\mu_k = \frac{\sum_{i=1}^m y_k^i x^i}{\sum_{i=1}^m y_k^i}$$

If label is given....Gaussian Naive Bayes!

$$\log L(\theta) = \sum_{i=1}^m \sum_{j=1}^k y_j^i \log \pi_j - \sum_{i=1}^m \log Z - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^k y_j^i (x^i - \mu_j)^\top \Sigma_j^{-1} (x^i - \mu_j)$$

$$\frac{\partial \log L}{\partial \mu_k} = - \sum_{i=1}^m y_k^i \Sigma_k^{-1} (x^i - \mu_k) = 0$$

$$\mu_k = \frac{\sum_{i=1}^m y_k^i x^i}{\sum_{i=1}^m y_k^i}$$

$$\frac{\partial \log L}{\partial \Sigma_k^{-1}} = - \sum_{i=1}^m y_k^i \left[\frac{\partial \log Z_k}{\partial \Sigma_k^{-1}} - \frac{1}{2} (x^i - \mu_k)(x^i - \mu_k)^\top \right] = 0$$

$$\pi_k = \frac{\sum_{i=1}^m y_k^i}{m}$$

$$\Sigma_k = \frac{\sum_{i=1}^m y_k^i (x^i - \mu_k)(x^i - \mu_k)^\top}{\sum_{i=1}^m y_k^i}$$

How to get label? Guess by current model

Gaussian Naive Bayes Prediction:

Compute the conditional probability by Bayes rule! $P(y|x) = \frac{P(y)\mathcal{N}(x|\mu_y, \Sigma_y)}{\sum_y P(y)\mathcal{N}(x|\mu_y, \Sigma_y)}$

$$y_j^l = \frac{\pi_l \mathcal{N}(x_j | \mu_l, \Sigma_l)}{\sum_{l=1}^k \pi_l \mathcal{N}(x_j | \mu_l, \Sigma_l)}$$

$$j = 1, \dots, m \qquad l = 1, \dots, k$$

Put everything together: Expectation-Maximization

For $t = 1, \dots$

- **E-Step**: Guess sample labels based on current model

$$y_j^l = \frac{\pi_l \mathcal{N}(x_j | \mu_l, \Sigma_l)}{\sum_{l=1}^k \pi_l \mathcal{N}(x_j | \mu_l, \Sigma_l)}$$

- **M-Step**: Update the parameters with current labels

$$\mu_k = \frac{\sum_{i=1}^m y_k^i x^i}{\sum_{i=1}^m y_k^i} \quad \pi_k = \frac{\sum_{i=1}^m y_k^i}{m} \quad \Sigma_k = \frac{\sum_{i=1}^m y_k^i (x^i - \mu_k) (x^i - \mu_k)^\top}{\sum_{i=1}^m y_k^i}$$

Expectation-Maximization

For $t = 1, \dots$

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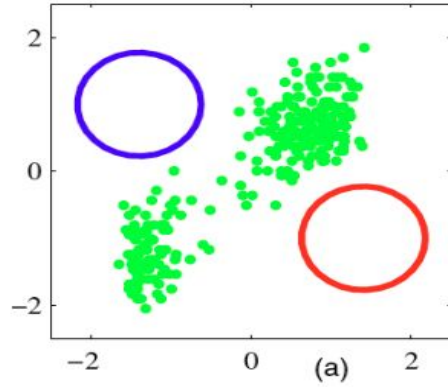
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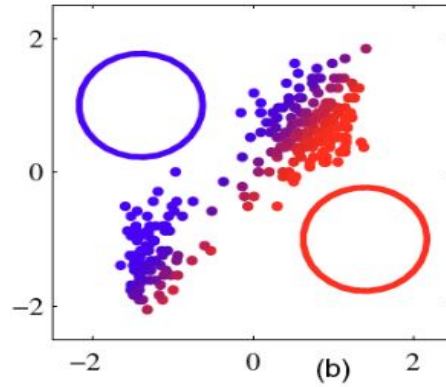
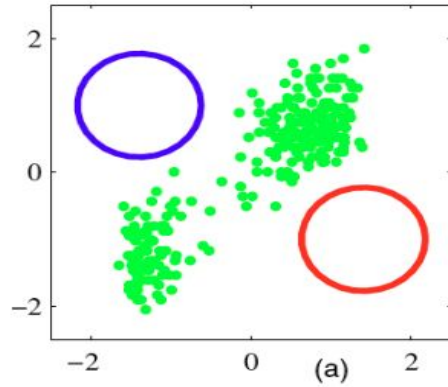
$$\mu_k = \frac{\sum_{i=1}^m y_k^i x^i}{\sum_{i=1}^m y_k^i} \quad \pi_k = \frac{\sum_{i=1}^m y_k^i}{m} \quad \Sigma_k = \frac{\sum_{i=1}^m y_k^i (x^i - \mu_k) (x^i - \mu_k)^\top}{\sum_{i=1}^m y_k^i}$$

This procedure is actually optimizing an upper bound of MLE, therefore, it converges

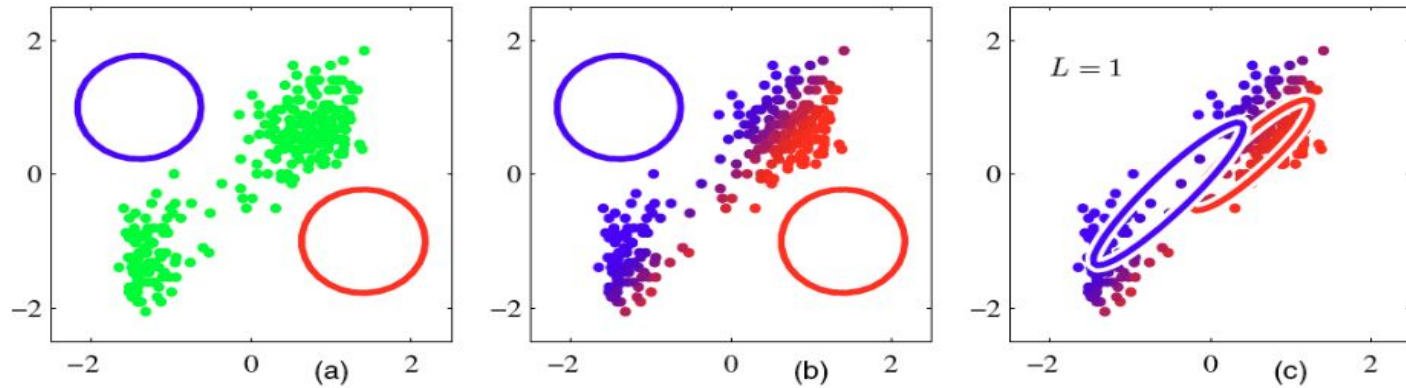
EM for GMM



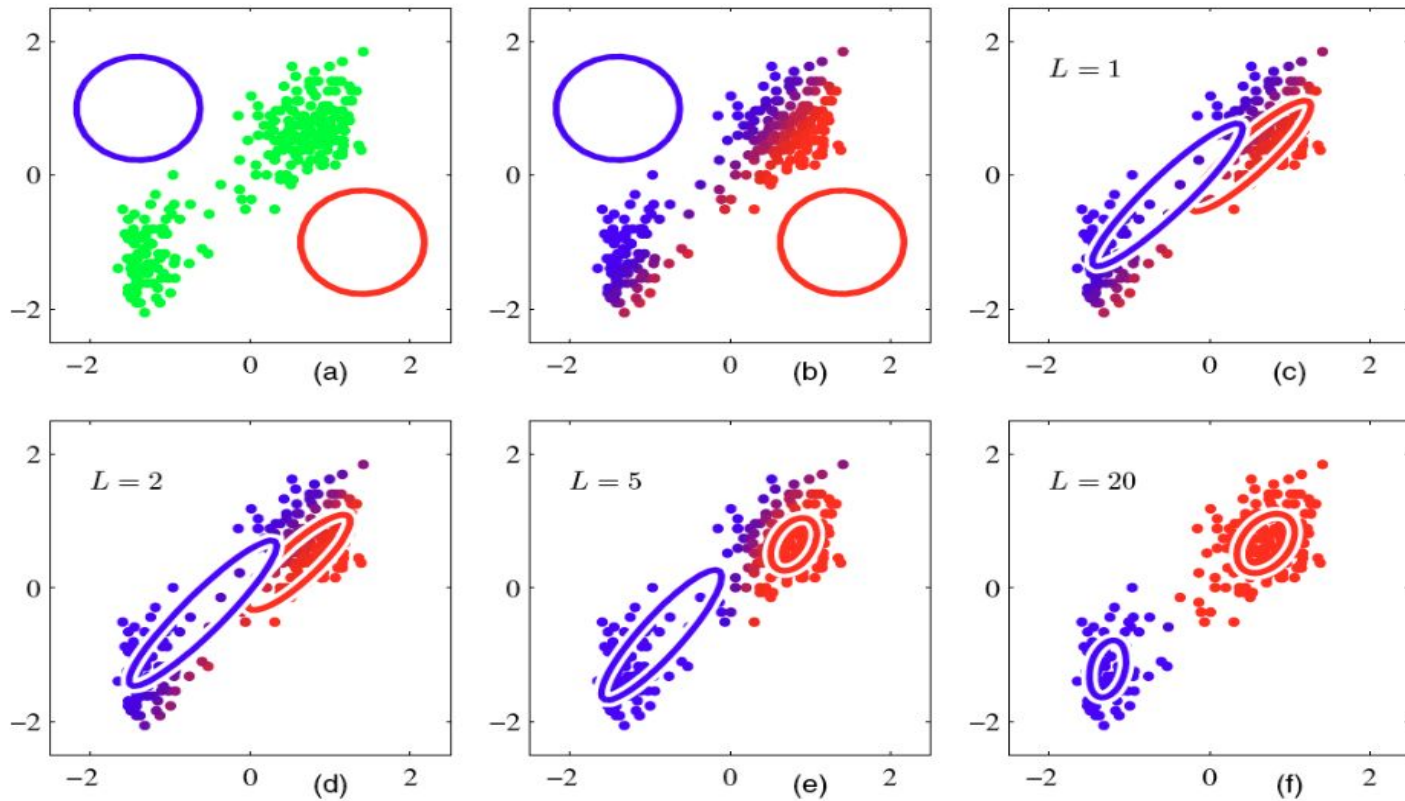
EM for GMM



EM for GMM



EM for GMM



Density Estimation: Gaussian Mixture Model



Density Estimation Pipeline

1. Build probabilistic models
Gaussian Mixture Model
2. Derive loss function (by MLE or MAP....)
MLE
3. Select optimizer
EM

Summary

GMM is a special case of hidden variable model

$$P(x) = \sum_y P(x|y)P(y)$$

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GMM is a special case of hidden variable model $P(x) = \sum_y P(x|y)P(y)$

A way of maximizing likelihood function for hidden variable models. It can be broken up into two (easy) pieces:

- Estimate some “missing” or “unobserved” data from observed data and current parameters.
- Using this “complete” data, find the maximum likelihood parameter estimates.

Summary

GMM is a special case of hidden variable model $P(x) = \sum_y P(x|y)P(y)$

A way of maximizing likelihood function for hidden variable models. It can be broken up into two (easy) pieces:

- Estimate some “missing” or “unobserved” data from observed data and current parameters.
- Using this “complete” data, find the maximum likelihood parameter estimates.

EM can converge, but can also get stuck in local minima.

Q&A